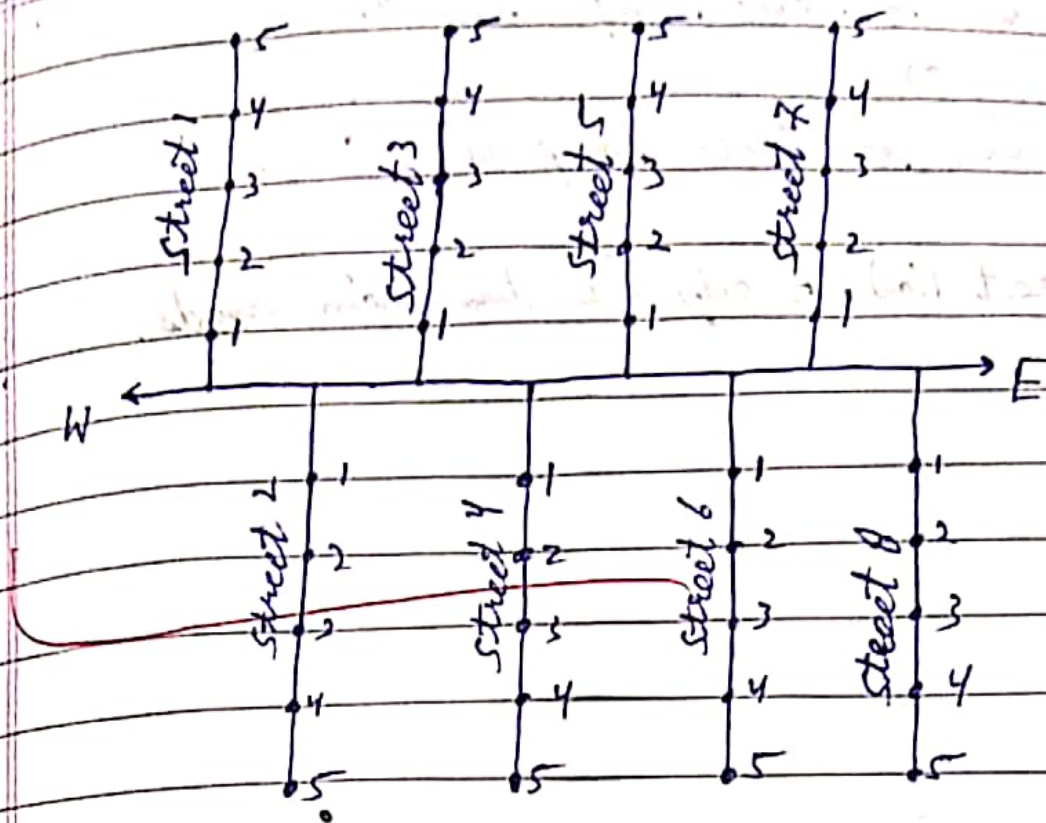


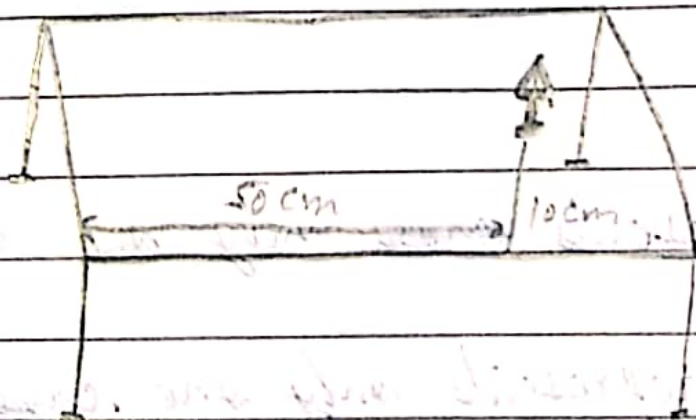
Coordinate Geometry.



Exercise: 4.1

Q1: How will you describe the position of a table lamp on your study table to another person?

Sol

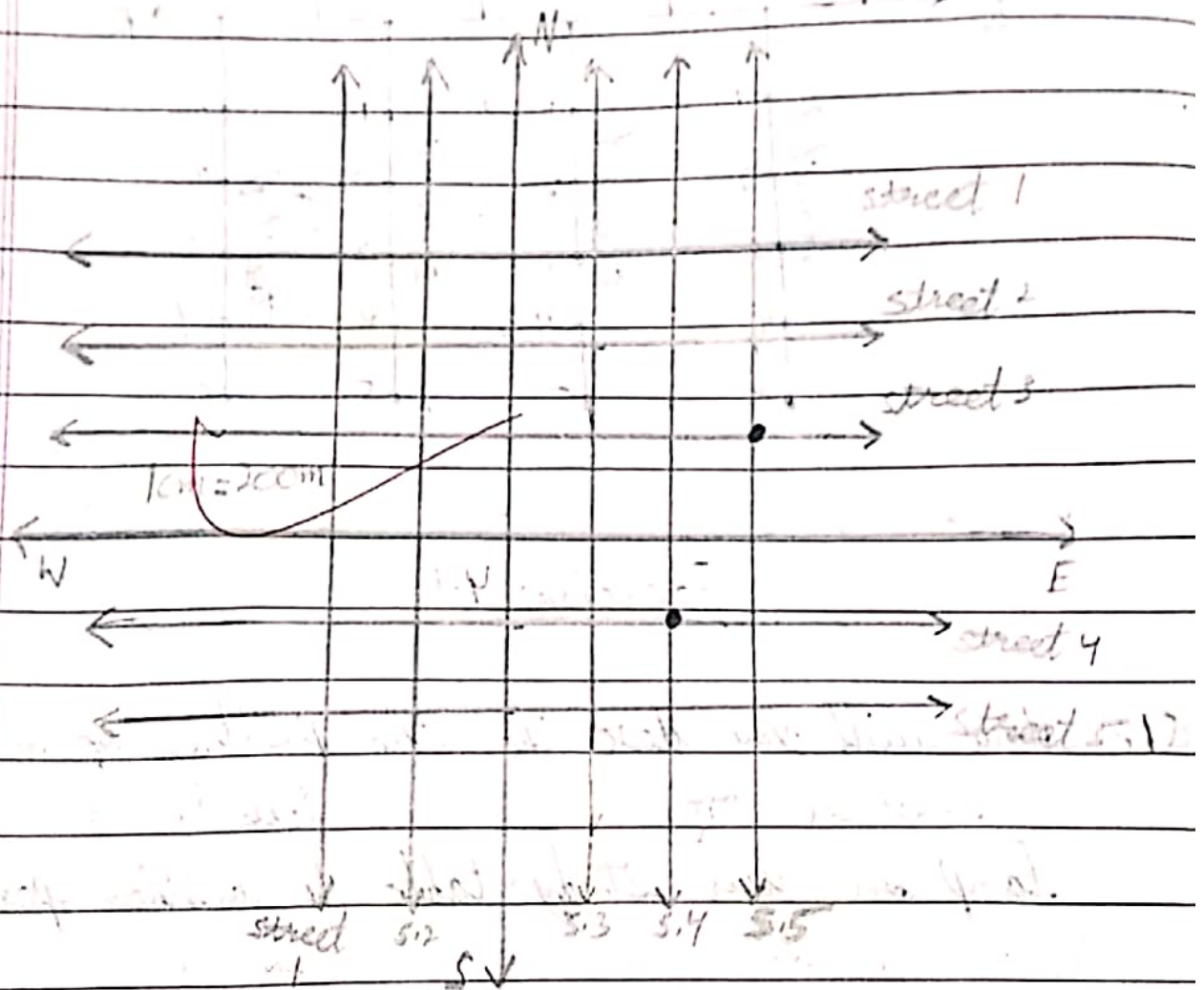


- Position of table lamp is $(10, 50)$.

or
Position of table lamp is $(50, 10)$.

Q2) (street Play): A city has two main roads.?

Sol:-

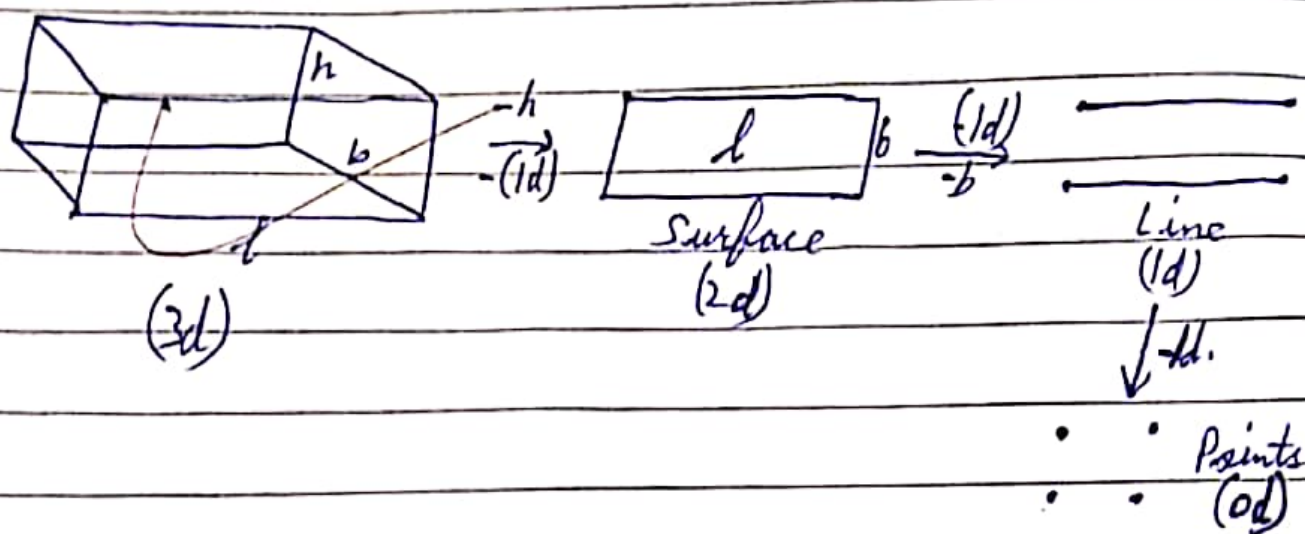


∴ (ii) $(4, 3)$ represents ~~cross~~ only one cross street.

ii) $(3, 4)$ also represents only one cross street.

⇒ Euclid's Geometry.

Ges - metry ~ A To measure the Earth.
↓ ↓
Earth - metrum → To measure.
(Greek word).



- Solid ~ A solid has a shape, position, volume and can be moved from one place to another. It's boundaries are called surface.
- They separate one part of space from another, and are said to have no thickness.

Page No. _____
Date ____/____/____

The boundaries of the surface are ~~some~~ curves or straight lines, these lines end in points.

⇒ Postulate:- Bases for a theory, argument or calculation is called a Postulate.

⇒ Axiom:- It is a statement or idea which people accept as being true.

⇒ Notation:- Idea or belief about something.

Elements

⇒ Some ~~Postulates~~ of Euclid:-

1) A point is that which has no part.

2) A line is breadthless length.

3) The ends of a line are points.

- 4) A straight line is a line which lies evenly with the point on itself.
- 5) A surface is that which has length and breadth only.
- 6) The edges of a surface are lines.
- 7) A plane surface is a surface which lies evenly with the straight lines on itself.

⇒ Euclid's Axioms:

Things which are equal to the same things are equal to one another.

If equals are added to equals, the wholes are equal.

Page No. _____
Date ____/____/____

3) If equals are subtracted from equals, the remainders are equal.

4) Things which coincide with one another are equal to one another.

5) The whole is greater than the part.

6) Things which are double of the same things are equal to one another.

7) Things which are halves of the same things are equal to one another.

→ Postulate 1: A straight line may be drawn from any one point to any other point.

• Axiom - 5.1: Given two distinct points, there is

a unique line that passes through them.

EXERCISE - 1

Q1: (i) Only one line can pass through a single Pt.

sol False, because through a single point there can pass infinitely many lines.

(ii). There are infinite number of lines passing through two distinct points.

sol False, because there passes only one line through two points.

(i) A terminated line can be produced indefinitely on both sides.

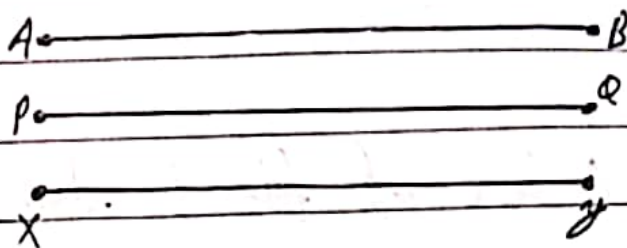
True: 

Page No. _____
Date ____/____/____

(iv) If two circles are equal then their radii are equal.

Sol. True.

(v) If $AB = PQ$
and $PQ = xy$
then $AB = xy$

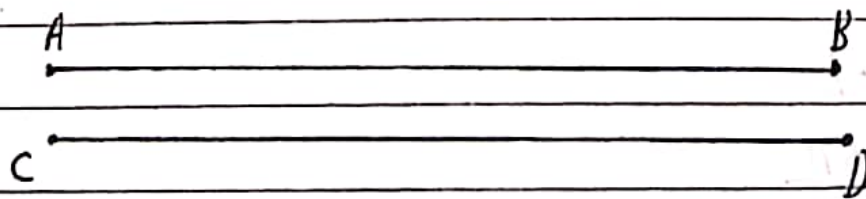


Sol True, $AB = PQ = xy$

$\Rightarrow AB = xy$

Q2. (i) Parallel Lines:-

Lines which do not intersect each other are called parallel lines.



ii) Perpendicular Lines:- Lines which make right

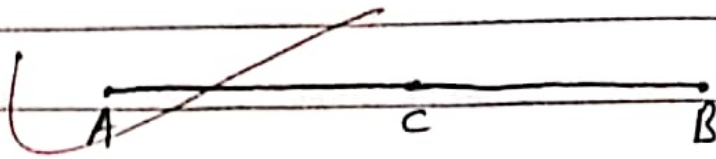
angle with one another are called perpendicular line.

→ Here we have to define 'Line' and 'angle' first.

(iii) Line Segment.

• A line of fixed dimension (Length) is called line segment.

Q 4.1



$$AB = AC + CB$$

$$= AC + BC$$

$$= AC + AC$$

$$= 2AC$$

$$\left(\begin{array}{l} AC = BC \\ BC = AC \end{array} \right)$$

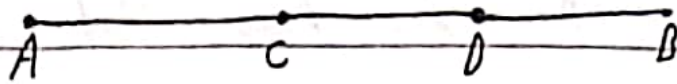
$$2AC = AB$$

$$AC = \frac{AB}{2}$$

$$AC = \frac{1}{2} AB$$

Q5. Prove that every line segment has one and only one mid point.

Sol.



Let C and D are two mid point's of AB

$$AB = \frac{1}{2} AC$$

$$AB = \frac{1}{2} AC, \text{ As C is mid point. } \dots \text{--- (1)}$$

$$AB = \frac{1}{2} AD, \text{ as D is the mid-point } \dots \text{--- (2)}$$

From (1) and (2), we have

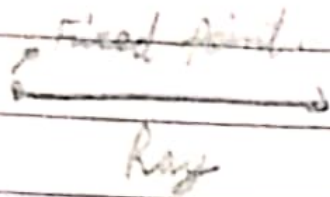
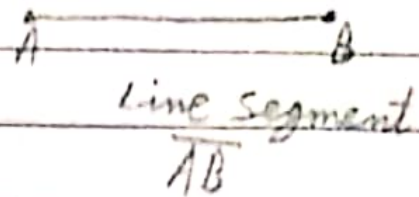
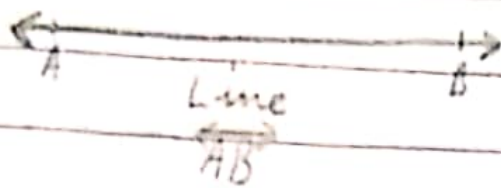
$$\frac{1}{2} AC = \frac{1}{2} AD$$

$$AC = \frac{AD}{2}$$

$AC = AD$, where which is a contradiction as AC is less than AD.

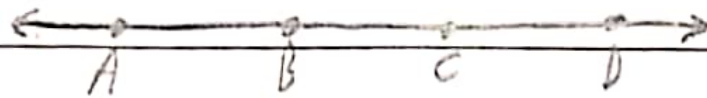
Lines And Angles

Page No. _____
Date ____/____/____

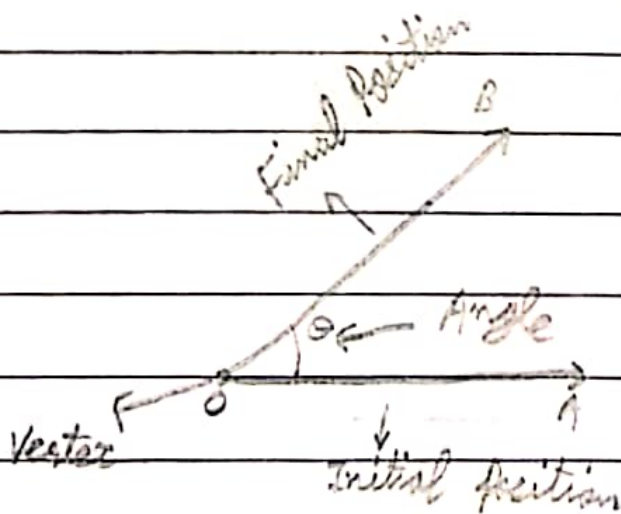


→ Collinear Points - Points on the same ~~side~~ line are called collinear points.

e.g.



(~~A, B, C, D~~ are collinear point)

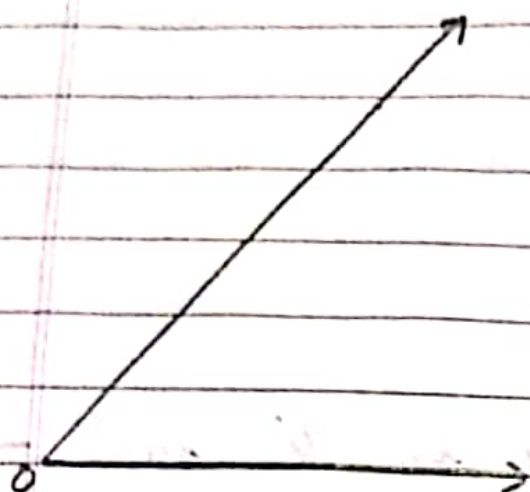


Arms of angle:
(OA and OB)

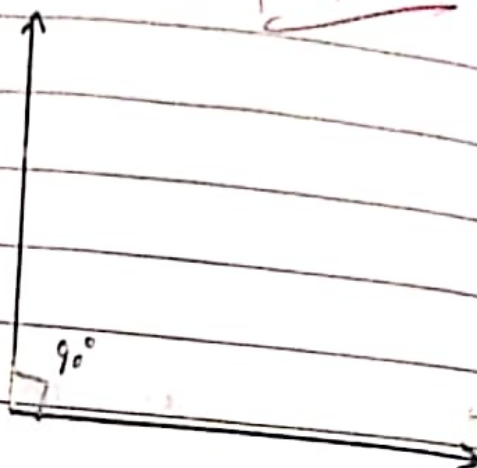
→ Angle - Angle is defined as the measurement of

rotation of a ray from its initial position

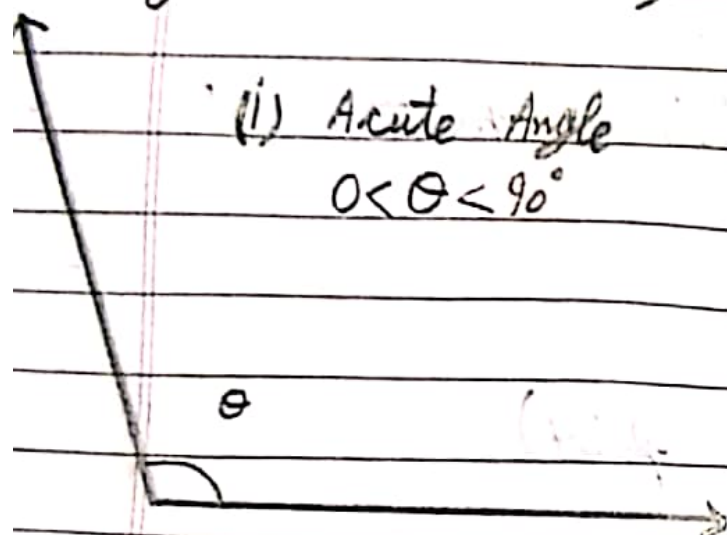
Q. ~~Q. No.~~
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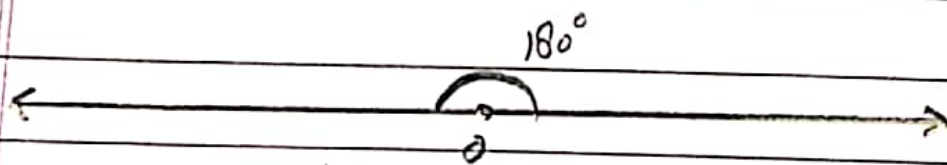
(i) Acute Angle
 $0 < \theta < 90^\circ$



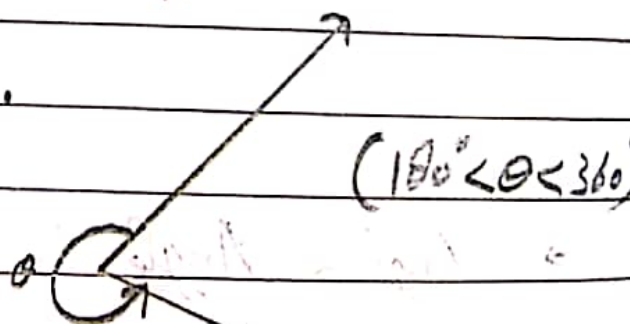
(ii) Right Angle (90°)



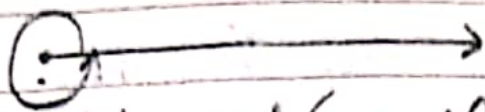
(iii) Obtuse Angle ($90^\circ < \theta < 180^\circ$)



(iv) Straight Angle.



(v) Reflex Angle



(vi) 360° (Complete Angle).

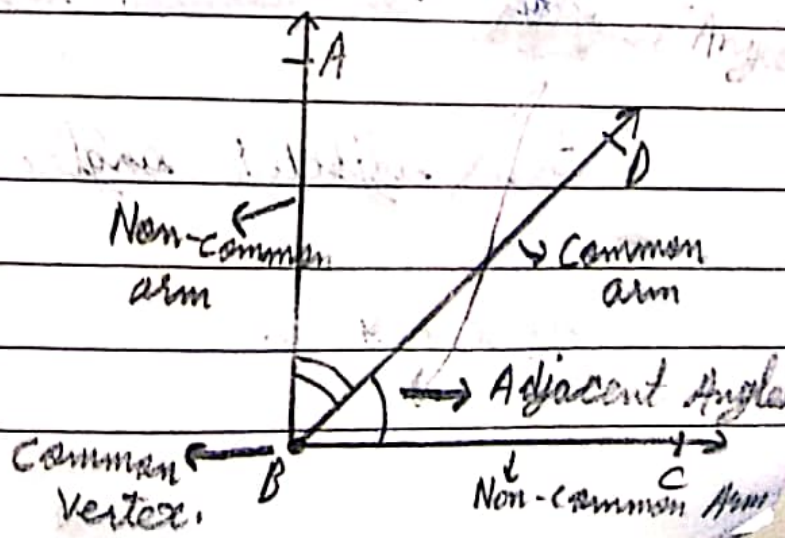
$\Rightarrow$ Complementary Angles:- If sum of two angles is 90° , then they are called Complementary angles.

e.g. $(60^\circ, 30^\circ) \rightarrow 60^\circ + 30^\circ = 90^\circ$

$\Rightarrow$ Supplementary Angles:- If sum of two angles is 180° , then they are called Supplementary angles.

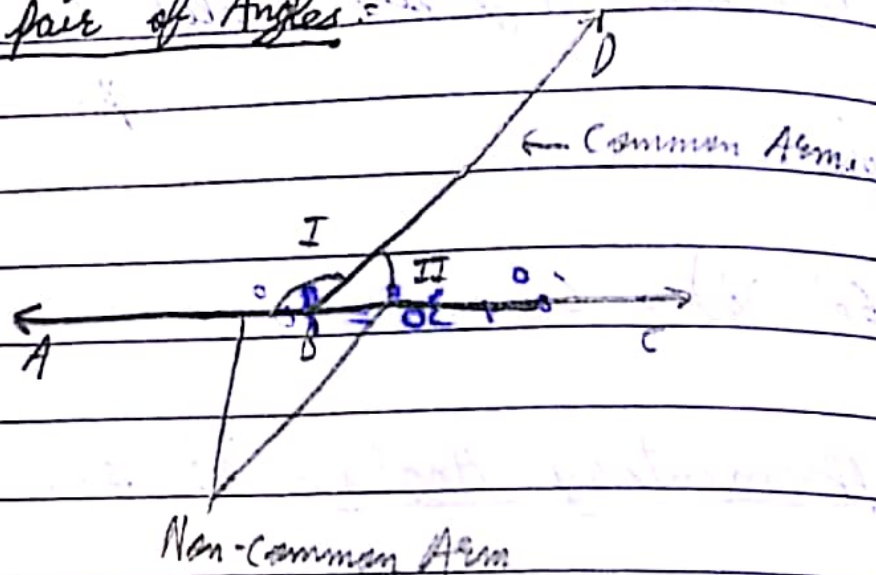
e.g. $(120^\circ, 60^\circ) \rightarrow 120^\circ + 60^\circ = 180^\circ$.

$\Rightarrow$ Adjacent Angles:-



- Two Angles are adjacent if they have a common vertex, a common Arm and their non-common arms are on different sides of the common arm.

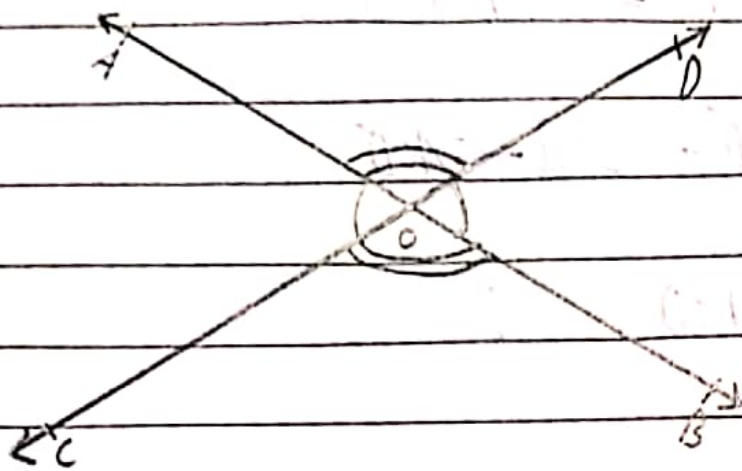
⇒ Linear pair of Angles:



- $\angle I$ and $\angle II$ are linear pair of angles.

When non-common arms form a straight line then adjacent angles are called Linear pair of angles.

$\Rightarrow$ Theorem 5.2:- If two lines intersect each other, then the vertically opposite angles are equal.



- Given:- Two lines AB and CD, intersecting each other at point O.
- To prove:- Vertically opposite angles are equal.
 i.e; (i) $\angle AOC = \angle BOD$
 (ii) $\angle AOD = \angle BOC$
- Proof:- Since $\angle AOC + \angle AOD = 180^\circ$ (L.P). --- (1)
- Again AOB is a straight line.

Page No. _____
Date: / / 20

$$\Rightarrow \angle AOD + \angle DOB = 180^\circ \text{ (L.P)} \dots (2).$$

- From (1) and (2), we have

$$\angle AOC + \angle AOD = \angle AOD + \angle BOD$$

$$\Rightarrow \angle AOC = \cancel{\angle AOD} + \angle BOD - \cancel{\angle AOD}$$

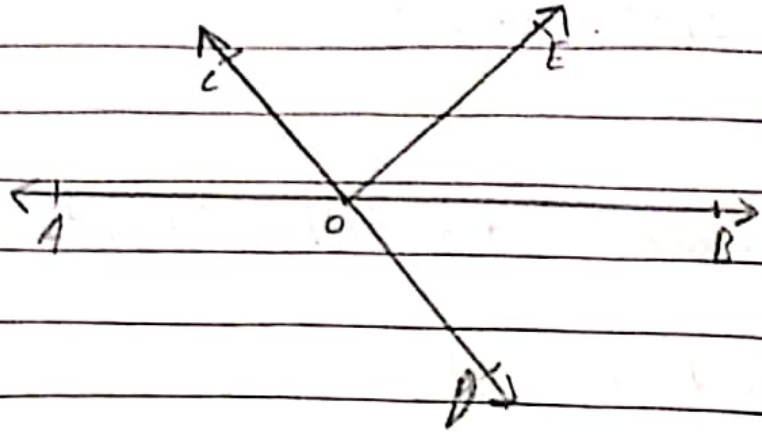
$$\Rightarrow \angle AOC = \angle BOD$$

EXERCISE - 5.1

Q1) In Fig, lines AB and CD intersect at O. If

$\angle AOC + \angle BOE = 70^\circ$ and $\angle BOD = 40^\circ$. Find $\angle BOE$ and

reflex $\angle COE$.



Sol: Given

$$\angle AOC + \angle BOE = 70^\circ$$

$$\angle BOD = 40^\circ$$

- Since $\angle AOC + \angle COE + \angle EOB = 180^\circ$

$$\Rightarrow (\angle AOC + \angle BOE) + \angle COE = 180^\circ$$

- $70^\circ + \angle COE = 180^\circ$

$$\angle COE = 180^\circ - 70^\circ$$

$$= 110^\circ$$

- Reflex $\angle COE = 360^\circ - \angle COE$

$$= 360^\circ - 110^\circ$$

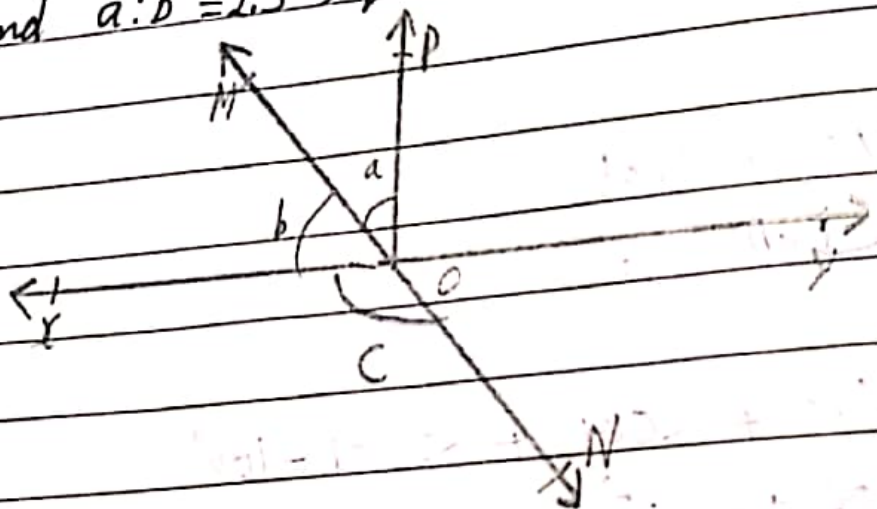
$$= 250^\circ$$

$\Rightarrow$ Now COD is a straight line.

$$\begin{aligned} \Rightarrow \angle COE + \angle BOE + \angle BOD &= 180^\circ \\ \Rightarrow 110^\circ + \angle BOE + 40^\circ &= 180^\circ \\ \Rightarrow 150^\circ + \angle BOE &= 180^\circ \\ \Rightarrow \angle BOE &= 180^\circ - 150^\circ \\ &= 30^\circ \end{aligned}$$

Q2: Lines XY and MN intersect at O. If $\angle POY = 90^\circ$

and $a:b = 2:3$, find C.



Sol Given,

$$\angle POY = 90^\circ \text{ and}$$

$$a:b = 2:3, C = ?$$

Since, $a:b = 2:3$

$$\Rightarrow \frac{a}{b} = \frac{2}{3}$$

$\Rightarrow a = 2k$
 $b = 3k$, (k is a natural number).

- Now, $\angle XOP + \angle POY = 180^\circ$ (Linear pair)

$\Rightarrow b + a + 90^\circ = 180^\circ$
 $b + a = 180^\circ - 90^\circ$
 $= 90^\circ$

- $3k + 2k = 90^\circ$
 $5k = 90^\circ$
 $k = \frac{90^\circ}{5}$

$k = 18$

$\therefore a = 2k = 2 \times 18 = 36^\circ$
 $b = 3k = 3 \times 18 = 54^\circ$

- Now, ~~the~~

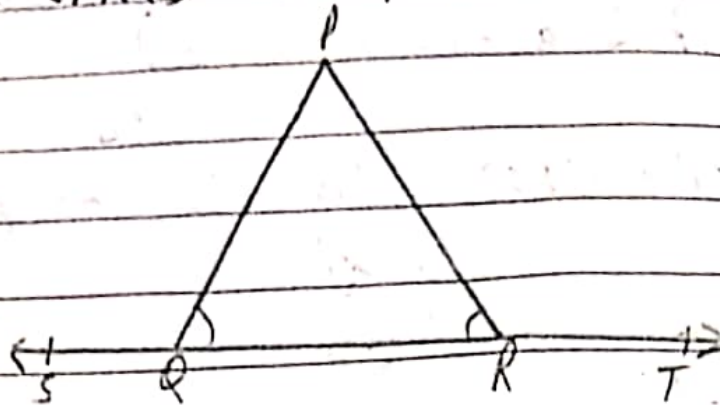
$b + c = 180^\circ$

$\Rightarrow 54^\circ + c = 180^\circ$

$c = 180^\circ - 54^\circ$

$c = 126^\circ$

Q3: $\angle PQR = \angle PRQ$, then prove that $\angle PQS = \angle PRT$.



Sol Given $\angle PQR = \angle PRQ$

- To prove: $\angle PQS = \angle PRT$

- Proof:-

Since $\angle PQS + \angle PQR = 180^\circ$ --- (1) (L.P)

Also, $\angle PRT + \angle PRQ = 180^\circ$ --- (2) (L.P)

From (1) and (2)

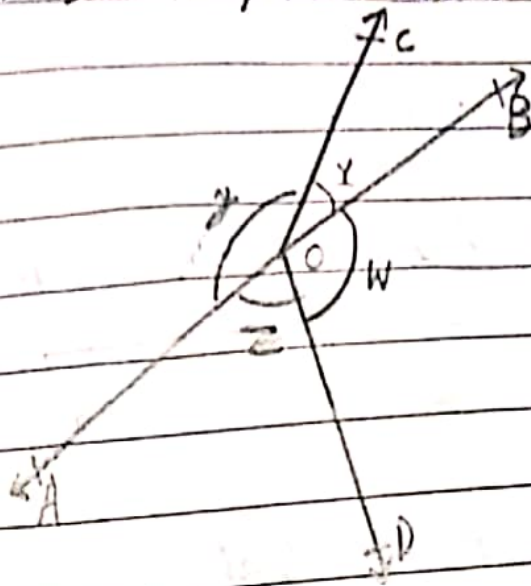
$$\angle PQS + \angle PQR = \angle PRT + \angle PRQ \quad (\because \angle PQR = \angle PRQ)$$

$$\Rightarrow \angle PQS + \angle PRQ = \angle PRT + \angle PRQ$$

$$\angle PQS = \angle PRT + \angle PRQ - \angle PRQ$$

$$\angle PQS = \angle PRT$$

Q4:- If $x+y = w+z$, then prove that AOB is a line.



Sol Given, $x+y = w+z$ ---- (1)

To prove: AOB is a line.

Since $x+y+z+w = 360^\circ$

$$\Rightarrow (x+y) + (w+z) = 360^\circ$$

$$(x+y) + (x+y) = 360^\circ \quad (\because w+z = x+y)$$

$$2x + 2y = 360^\circ$$

$$2(x+y) = 360^\circ$$

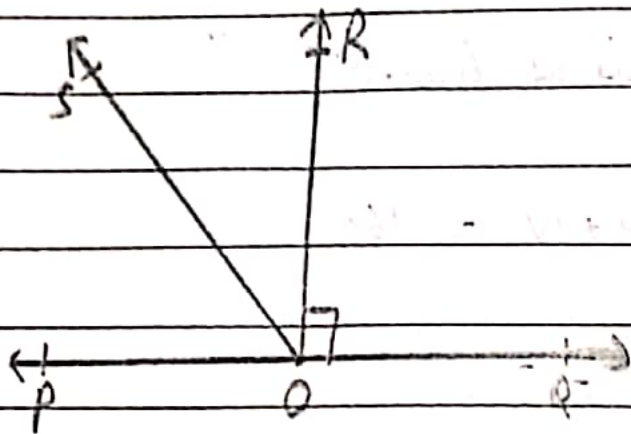
$$x+y = \frac{360^\circ}{2}$$

$$x + y = 180^\circ$$

$\Rightarrow AOB$ is a st. line.

Q5: ~~POQ~~ POQ is a line. Ray OR is perpendicular to line PQ. OS is another ray lying between ray OP and OR. Prove that

$$\angle ROS = \frac{1}{2} (\angle QOS - \angle POS)$$



sh Given:- POQ is a line. $OR \perp PQ$

- To prove:- $\angle ROS = \frac{1}{2} (\angle QOS - \angle POS)$

- Proof:-

Since $\angle ROQ + \angle POR = 180^\circ$ (L.P.)

$$\Rightarrow 90^\circ + \angle ROS + \angle POS = 180^\circ$$

$$[\angle POR = \angle POS + \angle ROS]$$

$$\therefore \angle ROS = 180^\circ - 90^\circ - \angle POS$$

$$\therefore \angle ROS = 90^\circ - \angle POS \quad \dots (1)$$

$$\Rightarrow \text{Now, } \angle QOS + \angle POS = 180^\circ$$

$$\Rightarrow \angle QOS + \angle POS - 2\angle POS = 180^\circ - 2\angle POS$$

$$\Rightarrow \angle QOS + \angle POS (1-2) = 2(90^\circ - \angle POS)$$

$$\Rightarrow \angle QOS + \angle POS \times -1 = 2(90^\circ - \angle POS)$$

$$\Rightarrow 2(90^\circ - \angle POS) = \angle QOS - \angle POS$$

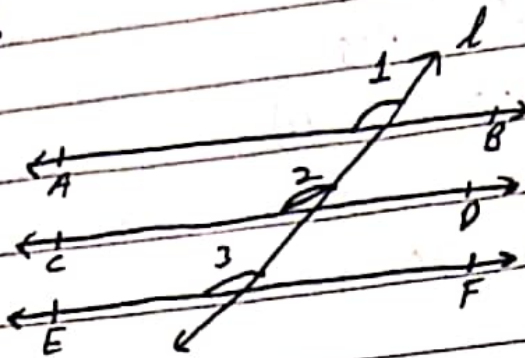
$$\Rightarrow 90 - \angle POS = \frac{\angle QOS - \angle POS}{2}$$

$$\Rightarrow \angle ROS = \frac{1}{2} (\angle QOS - \angle POS) \quad (\because \text{by 1})$$

~~Q6: It is~~

Theorem ^(5.3) Lines which are parallel to the same line are parallel to each other.

→ When lines are parallel, then corresponding angles are equal.



Given: $AB \parallel EF$ - $CD \parallel EF$

To prove: $AB \parallel CD$

Proof - Let l be the transversal of given lines

Since $AB \parallel EF$

$$\angle 1 = \angle 3 \text{ --- (1)}$$

$CD \parallel EF$

$$\angle 2 = \angle 3 \text{ --- (2)}$$

From (1) and (2)

We have.

$$\angle 1 = \angle 2$$

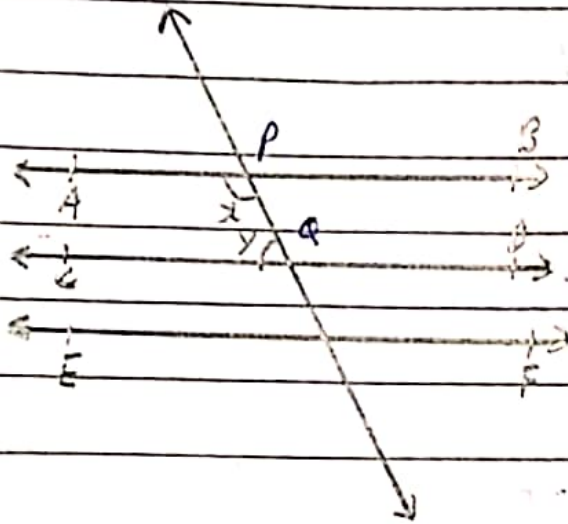
$\therefore$ In lines AB and CD

$$\angle 1 = \angle 2 \text{ (Proved)}$$

$\Rightarrow AB \parallel CD$ ($\because$ Corresponding Angles are equal).

Exercise: 5.2

Q1) If $AB \parallel CD$, $CD \parallel EF$ and $y:z = 3:7$, find x .



Sol Given,

$AB \parallel CD$,

$CD \parallel EF$ and

$$y:z = 3:7$$

- Since $\angle CQP = \angle RQD = y$ (Vertically opposite Angles)

Also, $y:z = 3:7$

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Page No. _____
Date ____/____/____ 12th

$$y = 3k, z = 7k$$

$$\text{Now, } y + z = 180^\circ$$

$$\Rightarrow 3k + 7k = 180^\circ$$

$$\Rightarrow 10k = 180^\circ$$

$$\Rightarrow k = \frac{180}{10}$$

$$k = 18$$

$$\therefore y = 3k = 3 \times 18 = 54^\circ$$

$$\therefore z = 7k = 7 \times 18 = 126^\circ$$

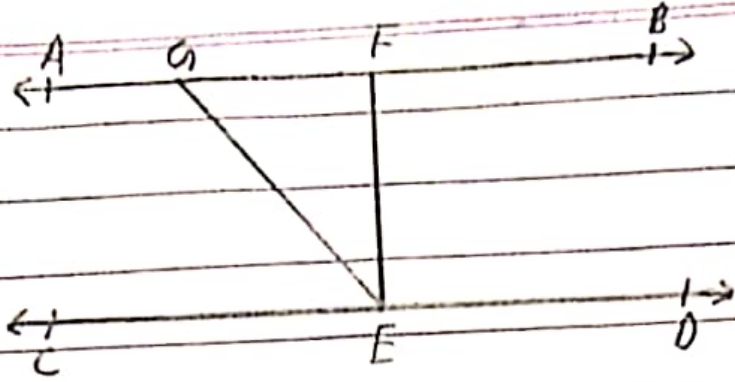
$$\Rightarrow AB \parallel CD, CD \parallel EF$$

$$\Rightarrow AB \parallel EF$$

$$\therefore x = z$$

$$x = 126^\circ \text{ (Alternate angles)}$$

Q2. If $AB \parallel CD$, $EF \perp CD$ and $\angle GED = 126^\circ$, find $\angle GEF$ and $\angle FGE$.



Sol Given, $AB \parallel CD$, $EF \perp CD$

$$\angle GED = 126^\circ$$

$$\Rightarrow AB \parallel CD$$

$$\angle AGE = \angle GED \text{ (Alternate Angles)}$$

$$= 126^\circ$$

$$\Rightarrow \angle AGE = 126^\circ$$

$$\angle GED = 126^\circ$$

$$\angle GEF + \angle FED = 126^\circ$$

$$\Rightarrow \angle GEF + 90^\circ = 126^\circ$$

$$\angle GEF = 126^\circ - 90^\circ$$

$$= 36^\circ$$

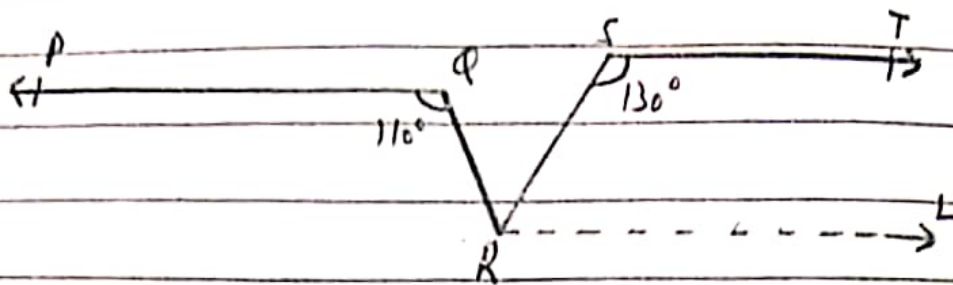
Now, $\angle FGE + \angle AGE = 180^\circ$

$$\Rightarrow \angle FGE + 126^\circ = 180^\circ$$

$$\Rightarrow \angle FGE = 180^\circ - 126^\circ$$

$$= 54^\circ$$

Q3: If $PQ \parallel ST$, $\angle PQR = 110^\circ$ and $\angle RST = 130^\circ$, Find $\angle QRS$.



Sol. Given: $PQ \parallel ST$,
 $\angle PQR = 110^\circ$
 $\angle RST = 130^\circ$

$\Rightarrow$ Construction: Draw $RL \parallel ST$

$\therefore PQ \parallel ST, RL \parallel ST$

$\Rightarrow PQ \parallel RL$

$$\angle PQR = \angle QRL$$

$$\Rightarrow 110^\circ = \angle QRL \Rightarrow \angle QRL = 110^\circ$$

Again $ST \parallel RL$

$$\angle RST + \angle SRL = 180^\circ$$

$$130^\circ + \angle SRL = 180^\circ$$

$$\begin{aligned} \angle SRL &= 180^\circ - 130^\circ \\ &= 50^\circ \end{aligned}$$

Now, $\angle QRS + \angle 50^\circ = 110^\circ$

$\Rightarrow \angle QRS = 110^\circ - 50^\circ$
 $= 60^\circ$